TECHNOLOGY-INTEGRATED METACOGNITIVE ARGUMENT-DRIVEN INQUIRY

(TI-MADI) APPROACH

CHEMISTRY 10 MODULE



NAME:

GR. & SEC:

INTRODUCTORY MESSAGE

Hello little Einsteins and Curies!

Welcome to the technology-integrated metacognitive argument-driven inquiry (TI-MADI) approach module in Chemistry 10. I am thrilled to launch this learning journey with you. The TI-MADI approach module is designed to improve the conceptual understanding, science process skills, and self-efficacy toward learning chemistry of Grade 10 students. In the duration of this course, you will acquire significant understanding pertaining to Boyle's Law, Charles' Law, Biomolecules, Chemical Equations, the Law of Conservation of Mass, and Factors influencing Chemical Reactions.

This module showcases the fusion of technology into the combination of interactive exercises, virtual simulations, and cooperative investigations to ensure your continuous engagement. You shall have a chance to put into practice the knowledge you acquire through sessions of argumentation. This module promotes cooperation and discourse. Do not hesitate to express your viewpoints and inquiries with your fellow peers.

Here are the key seven parts of each lesson in the TI-MADI approach module:

- **Refreshing Knowledge** – what you know about the topic
- **Stimulating Insights** – introduce new learning
- **Investigating Moments** – explore virtual simulations
- **Justifying Arguments** – communicate new findings
- **Expounding Era** – further discussion of the topic
- **Checking Compositions** – craft and assess written argumentations
- **Reflection** – reflect on the lesson

For any questions or concerns, please reach out to your instructor. Additionally, the Google Classroom has been set up for discussions and announcements. I am here to support you throughout this module. Your success is our priority as an educator, and I believe that with dedication and active participation, you will excel in the TI-MADI approach module.

Best Regards,
Author

TABLE OF CONTENTS

Title Page 1

Introductory Message 2

Getting Started 4

General Instructions 5

Lesson 1 Boyle's Law 6

Lesson 2 Charles' Law 13

Lesson 3 Biomolecules 20

Lesson 4 Law of Conservation of Mass 27

Lesson 5 Balancing Chemical Equations 34

Lesson 6 Factors affecting Reaction Rates 41

GETTING STARTED

Overview

This TI-MADI approach module will help you understand the concepts of chemistry in Grade 10. At the end of the module, you are expected to accomplish the following objectives per topic:

Boyle's law

- Perform an experiment about Boyle's Law using a virtual simulation;
- Solve mathematical problems using the Boyle's Law equation; and
- Write on the reflective log the importance of Boyle's Law.

Charles' law

- Demonstrate the relationship between volume and temperature using a virtual simulation;
- Calculate the missing variables of the given mathematical problems related to Charles' Law; and
- Reflect upon the importance of Charles' Law in real-life situations such as lung capacity during winter.

Biomolecules

- Classify the major types of biomolecules including carbohydrates, lipids, proteins, and nucleic acid;
- Investigate the structure of DNA through an interactive activity; and
- Internalize the knowledge about biomolecules in real-life situations such as meal planning.

Law of Conservation of Mass

- Define the Law of Conservation of Mass;
- Investigate upon the Law of Conservation of Mass using a virtual simulation; and
- Write reflection about the importance of chemical reactions in practical scenarios such as
- combustion.

Balancing Chemical Equations

- Explain the factors required for a chemical equation to be balanced;
- Perform an investigation about chemical reactions using a virtual simulation; and
- Summarize the importance of balancing chemical equations.

Factors affecting chemical reactions

- Describe each factors affecting chemical reactions
- Examine the concepts of collision theory using a virtual simulation;
- Reflect how the factors affecting rates of chemical reactions are utilized in food industry and materials production, control of fire, pollution, and corrosion;

GENERAL INSTRUCTIONS

1. Read the instructions carefully before performing each activity.

2. Follow the proper sequence of the included tasks.

3. Access the links by scanning QR codes or directly clicking on the links provided.

4. Save a screenshot of the outputs done in online applications or websites.

5. Dedicate enough time to read and comprehend the lesson. Adhere closely to the instructions and diligently complete all activities.

6. Maintain honesty throughout the completion of activities, accurately record only observations that have been truly made.

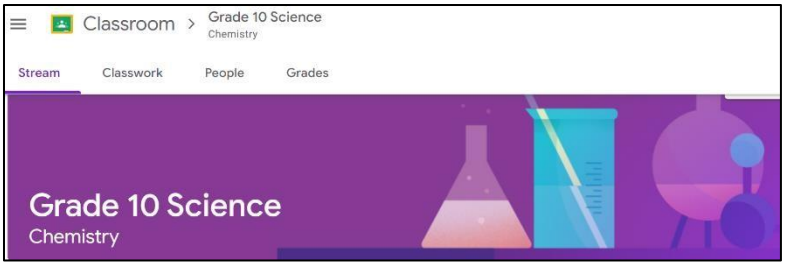
7. Always engage in cooperation with both the group leader and fellow members whilst participating in collaborative activities.

8. Do not hesitate to solicit clarification. If there is a necessity to enhance comprehension, approach your instructor. Furthermore, consult alternative sources for supplementary enlightenment.

LEARNING MANAGEMENT SYSTEM

Once you have received this module, you may join the Learning Management System of our class for this module by visiting the Google Classroom website and entering the code: azczxhu or clicking on this link: http://tinyurl.com/sci10gclass. You may also scan the QR code using a QR code scanner to access the G-classroom.

SCAN ME!



Interface of G-classroom

LESSON 1

BOYLE'S LAW

LEARNING OBJECTIVES:

At the end of the lesson, the learners must be able to:

- Perform an experiment about Boyle's Law using a virtual simulation;
- Solve mathematical problems using the Boyle's Law equation; and
- Write on the reflective log the importance of Boyle's Law.

DURATION:

4 Hours

INTRODUCTION:

Imagine a world where your breath could inflate a balloon to the size of a car, and a simple squeeze could turn it into a speck. Welcome to the intriguing realm of Boyle's Law, where the pressure and volume of gases dance in a captivating tango, revealing secrets that power engines, dictate the depths of underwater exploration, and even shape the air we breathe. In this journey, we'll unravel the mysteries of Boyle's Law, the scientific symphony that lets us compress, expand, and manipulate the invisible forces that surround us, all with a touch of curiosity and a dash of science.

CHEM-O-TRIVIA

Did you know? Boyle's Law is named after Robert Boyle, an Anglo-Irish natural philosopher, chemist, and physicist who first stated the law in 1662. However, the relationship between pressure and volume was also independently discovered by the French scientist Edme Mariotte around the same time.

REFRESHING KNOWLEDGE

DIRECTION: Using your web browser, visit the link to access the Word Wall website, or using a QR code scanner app, scan the QR code below. Click the start button and play the word search game. Then upload a screenshot or picture of your garnered score on the g-classroom "W1 Refreshing Knowledge" classwork.



Figure 1. *Interface of Word Search in Word Wall*

GUIDING QUESTION

In this lesson, you will be formulating a scientific argument to answer the guiding question:

If temperature is held constant, what is the relationship between volume and pressure?

PLANNING METACOGNITIVE CHECKPOINT

DIRECTION: Answer and brainstorm the following questions with your groupmates.

- What is the description of the task?
- Why this task is important to us?
- What is our main goal?
- What are the resources that we need to successfully finish this task?
- How much time will we need to finish this task?
- How do we divide the task?



INVESTIGATING MOMENTS

DIRECTION: Using your web browser, visit the link: https://tinyurl.com/ExplorationPhysClass to access The Physics Classroom, or using a QR code scanner application, scan the QR code below. Working collaboratively with your groupmates, feel free to explore the simulation to record your observations and gather relevant data that will serve as evidence in your initial scientific argument.

SCAN ME!

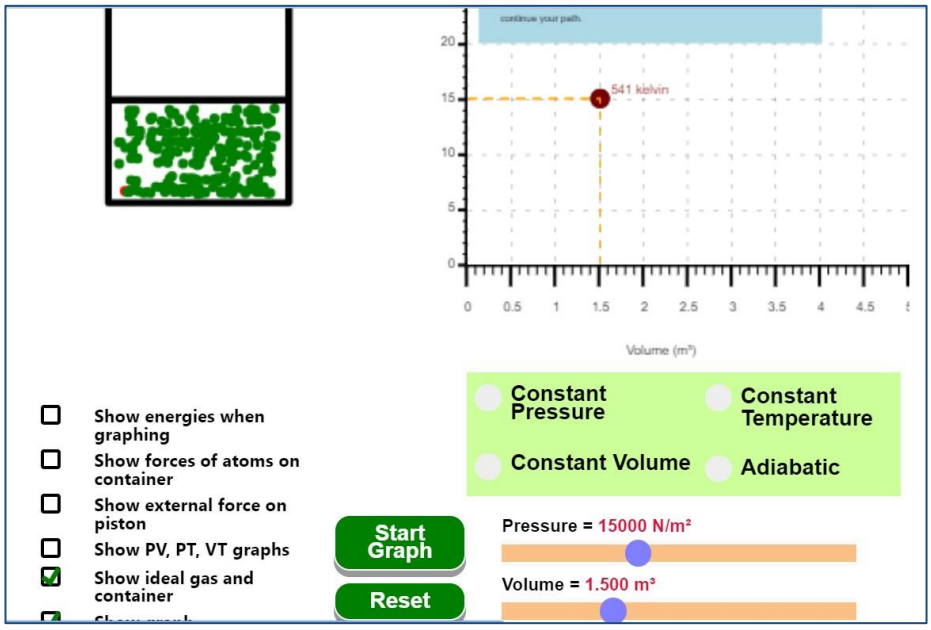


Figure 4. *Interface of The Physics Classroom Chemistry Simulation*

MONITORING METACOGNITIVE CHECKPOINT

DIRECTION: Answer and brainstorm the following questions with your groupmates.

- Do we understand what we are doing?
- Does the task make sense?
- Are we reaching our goals?
- Are the strategies working well to complete the task within the given time limit? If not, what should we do?
- Do we need to make improvements?



JUSTIFYING ARGUMENTS

DIRECTION: After exploring the simulation, finalize your work and data. Record your observations on the digital argumentation board using the template in Canva by clicking on this link or using a QR scanner application, scan the QR code. To answer the guiding question, you need to develop a scientific argument consisting of the following:

SCAN ME!

- **claim** (an explicit statement or conclusion which answers the guiding question),
- **evidence** (data observed and collected from the exploration that supports your claim), and
- **reasoning** (explains the importance of data presented to support your claim and connects related scientific principles to provide evidence to your claim).

To share your initial findings with others, we will employ a round-robin arrangement. This implies that two members from your group will remain stationary at your room table (RT) station to explain your group's work while the remaining members of your group will round to the remaining RT stations consecutively to attentively perceive and evaluate the initial findings formulated by your classmates.

Be ready to present your group's initial scientific argument. The aim of the argumentation session is not to convince others that your argument is the best, but the goal is to identify errors or faulty reasoning in the arguments and fix those mistakes. You will need to evaluate the content of the claim, the quality of the evidence, and the strength of the reasoning in each argument. To share your argument, you may raise your hand as your teacher facilitates the discussion.

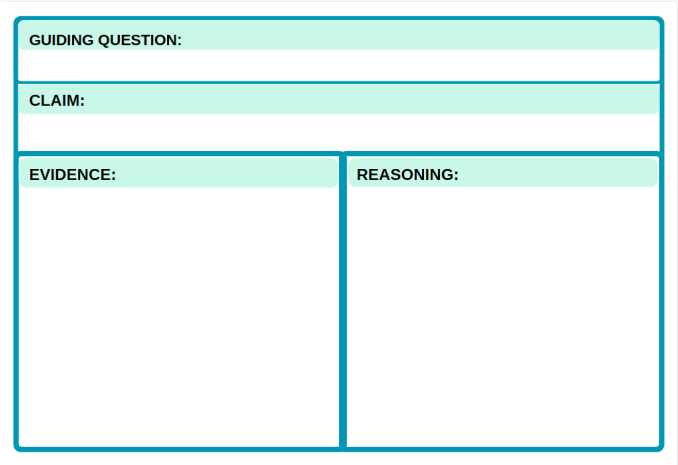


Figure 5. *Initial Argumentation Board*

EXPOUNDING ERA

DIRECTION: Using your web browser, visit the link: https://byjus.com/chemistry/boyles-law/ or using a QR scanner application, scan the QR code below for you to access BYJU'S website for further elaboration of the topic. Sample problem-solving items and real-life examples are also on the website.

SCAN ME!



Figure 6. *Interface of BYJU's Website*

EVALUATING METACOGNITIVE CHECKPOINT



DIRECTION: Answer and brainstorm the following questions with your groupmates.

- Have we reached our goal?
- What strategy worked?
- What strategy did not work?
- Would we do things differently next time?

CHECKING COMPOSITIONS

DIRECTION: Once you have completed your group investigation, participated in the argumentation session, and reflective post-discussion, you will need to finalize your individual written scientific argumentation report. Be reminded that your scientific argument must consist of a claim (a clear statement that answers the guiding question), **evidence** (data from the investigation that supports your claim), and **reasoning** (explains clearly why the data you presented supports your claim and consists of underlying scientific principles that link evidence to your

GUIDING QUESTION:	
CLAIM:	
EVIDENCE:	REASONING:

REFLECTION

DIRECTION: You may now answer the following reflective questions via Google Forms by clicking on this link: http://tinyurl.com/W1REFLECTIVELOG or scanning the QR code below.

SCAN ME!

1. Which helps you to learn more in this session?

   _____
   _____
   _____

2. What are the important details from the lesson you learned?

   _____
   _____
   _____

3. How will you apply your learnings in real-life situations?

   _____
   _____
   _____

4. What topic does not come easy in our class? Hint some suggestions to overcome those hardships.

   _____
   _____
   _____

LESSON 2

CHARLES' LAW

LEARNING OBJECTIVES:

At the end of the lesson, the learners must be able to:

- Demonstrate the relationship between volume and temperature using a virtual simulation;
- Calculate the missing variables of the given mathematical problems related to Charles' Law; and
- Reflect upon the importance of Charles' Law in real-life situations such as lung capacity during winter.

DURATION:

4 Hours

INTRODUCTION:

Have you ever enjoyed the aromatic smell of newly baked bread? The mixing of the right ingredients is vital to obtain the desired texture and flavor of bread. The fluffiness of bread depends on the amount of yeast added to it. The yeast feeds on sugar and converts it into carbon dioxide. When placed in an oven at a high temperature, the dough will rise. Discovered by French scientist Jacques Charles, we have applied in the previous example the Charles' Law that describes the relationship between the temperature and volume of a gas when pressure is kept constant. Understanding the concepts about Charles' Law will aid us in exploring the behaviors of gases in different scientific and industrial applications.

CHEM-O-TRIVIA

Did you know? Charles' Law is applied in different industries, including aerospace, chemistry, and food processing.

REFRESHING KNOWLEDGE

DIRECTION: Using your web browser, visit the link: http://tinyurl.com/W2RefreshingKnowledge to access the Word Wall website, or using a QR code scanner app, scan the QR code below. Click the start button and play the Match up game. Then upload a screenshot or picture of your garnered score on the g-classroom "W2 Refreshing Knowledge" classwork.

SCAN ME



Figure 1. *Interface of Match up in Word Wall*

GUIDING QUESTION

In this lesson, you will be formulating a scientific argument to answer the guiding question:

If pressure is held constant, what is the relationship between volume and temperature?

PLANNING METACOGNITIVE CHECKPOINT

DIRECTION: Answer and brainstorm the following questions with your groupmates.

- What is the description of the task?
- Why this task is important to us?
- What is our main goal?
- What are the resources that we need to successfully finish this task?
- How much time will we need to finish this task?
- How do we divide the task?



INVESTIGATING MOMENTS

DIRECTION: Using your web browser, visit the link: http://tinyurl.com/W2VirtualSimulation to access the PhET Virtual Simulation, or using a QR code scanner application, scan the QR code below. To start the simulation, click on the "Ideal" button. Working collaboratively with your groupmates, feel free to explore the simulation to record your observations and gather relevant data that will serve as evidence in your initial scientific argument.

SCAN ME

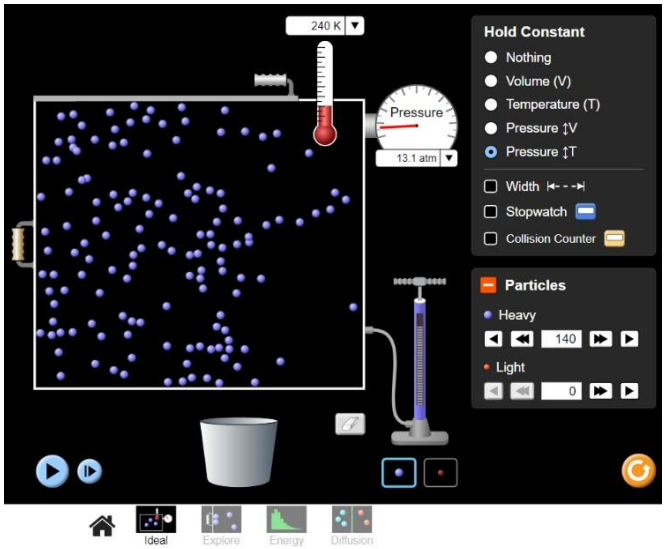


Figure 4. *Interface of the PhET Simulation*

MONITORING METACOGNITIVE CHECKPOINT

DIRECTION: Answer and brainstorm the following questions with your groupmates.

- Do we understand what we are doing?
- Does the task make sense?
- Are we reaching our goals?
- Are the strategies working well to complete the task within the given time limit? If not, what should we do?
- Do we need to make improvements?



JUSTIFYING ARGUMENTS

DIRECTION: After exploring the simulation, finalize your work and data. Record your observations on the digital argumentation board using the template in Canva by clicking on this link: http://tinyurl.com/W2ADIBoard or using a QR scanner application, scan the QR code. To answer the guiding question, you need to develop a scientific argument consisting of the following:

SCAN ME!

- **claim** (an explicit statement or conclusion which answers the guiding question),
- **evidence** (data observed and collected from the exploration that supports your claim), and
- **reasoning** (explains the importance of data presented to support your claim and connects related scientific principles to provide evidence to your claim).

To share your initial findings with others, we will employ a round-robin arrangement. This implies that two members from your group will remain stationary at your room table (RT) station to explain your group's work while the remaining members of your group will round to the remaining RT stations consecutively to attentively perceive and evaluate the initial findings formulated by your classmates.

Be ready to present your group's initial scientific argument. The aim of the argumentation session is not to convince others that your argument is the best, but the goal is to identify errors or faulty reasoning in the arguments and fix those mistakes. You will need to evaluate the content of the claim, the quality of the evidence, and the strength of the reasoning in each argument. To share your argument, you may raise your hand as your teacher facilitates the discussion.

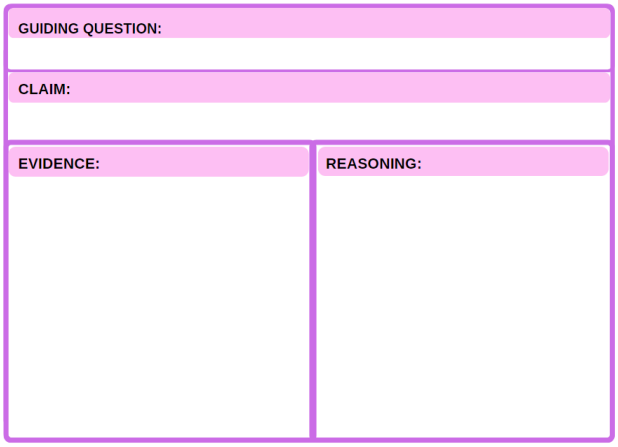


Figure 5. *Initial Argumentation Board*

EXPOUNDING ERA

DIRECTION: Using your web browser, visit the link: https://byjus.com/jee/charles-law/ or using a QR scanner application, scan the QR code below for you to access BYJU'S website for further elaboration of the topic. Sample problem-solving items and real-life examples are also on the website.

SCAN ME

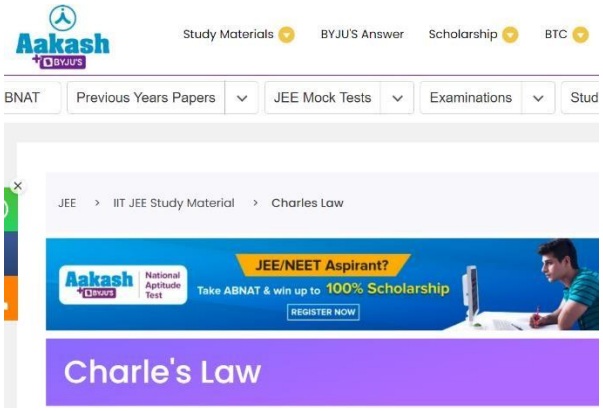


Figure 6. *Interface of BYJU's Website*

EVALUATING METACOGNITIVE CHECKPOINT



DIRECTION: Answer and brainstorm the following questions with your groupmates.

- Have we reached our goal?
- What strategy worked?
- What strategy did not work?
- Would we do things differently next time?

CHECKING COMPOSITIONS

DIRECTION: Once you have completed your group investigation, participated in the argumentation session, and reflective post-discussion, you will need to finalize your individual written scientific argumentation report. Be reminded that your scientific argument must consist of a claim (a clear statement that answers the guiding question), **evidence** (data from the investigation that supports your claim), and **reasoning** (explains clearly why the data you presented supports your claim and consists of underlying scientific principles that link evidence to your claim).

GUIDING QUESTION:	
CLAIM:	
EVIDENCE:	REASONING:

REFLECTIVE LOG

DIRECTION: You may now answer the following reflective questions via Google Forms by clicking on this link: http://tinyurl.com/W2REFLECTIVELOG or scanning the QR code below.

SCAN ME

1. What are the three important things that you learned in this session?

_____

_____

_____

2. What are the two things that you cannot forget?

_____

_____

_____

3. What is one thing that you want to share with others?

_____

_____

_____

LESSON 3

BIOMOLECULES

LEARNING OBJECTIVES:

At the end of the lesson, the learners must be able to:

- Classify the major types of biomolecules including carbohydrates, lipids, proteins, and nucleic acid;
- Investigate the structure of DNA through an interactive activity; and
- Internalize the knowledge about biomolecules in real-life situations such as meal planning.

DURATION:

4 Hours

INTRODUCTION:

The construction, operation, and regulation of living things depend on organic molecules known as biomolecules. These molecules form the foundation of life and are essential to many biological functions. Biomolecules are classified into four primary classes: proteins, carbohydrates, lipids, and nucleic acids. Biomolecules include both small primary and secondary metabolites and large macromolecules. Biomolecules are organic compounds, yet they are distinct from other organic molecules in that they have the same chemical kinds, behave similarly when they react, and adhere to the same physical laws.

CHEM-O-TRIVIA Did you know? Because it forms four bonds in a tetrahedral shape and can take the form of linear, branched, cyclic, or aromatic carbon skeletons, carbon is the most prevalent element in biomolecules and functions similarly to organic compounds.

REFRESHING KNOWLEDGE

DIRECTION: Using your web browser, visit the link: http://tinyurl.com/W3RefreshingKnowledge to access the Word Wall website, or using a QR code scanner app, scan the QR code below. Click the start button and play the Missing word game. Then upload a screenshot or picture of your garnered score on the g-classroom "W3 Refreshing Knowledge" classwork.

SCAN ME



Figure 1. *Interface of Missing word in Word Wall*

GUIDING QUESTION

In this lesson, you will be formulating a scientific argument to answer the guiding question:

How do the structures of carbohydrates differ from those of lipids, and what functions do these biomolecules serve in cells?

PLANNING METACOGNITIVE CHECKPOINT

DIRECTION: Answer and brainstorm the following questions with your groupmates.

- What is the description of the task?
- Why this task is important to us?
- What is our main goal?
- What are the resources that we need to successfully finish this task?
- How much time will we need to finish this task?
- How do we divide the task?



INVESTIGATING MOMENTS

DIRECTION: Using your web browser, visit the link: http://tinyurl.com/Lesson3Simulation to access BioMan Biology Virtual Simulation, or using a QR code scanner application, scan the QR code below. Working collaboratively with your groupmates, feel free to explore the simulation to record your observations and gather relevant data that will serve as evidence in your initial scientific argument.

SCAN ME

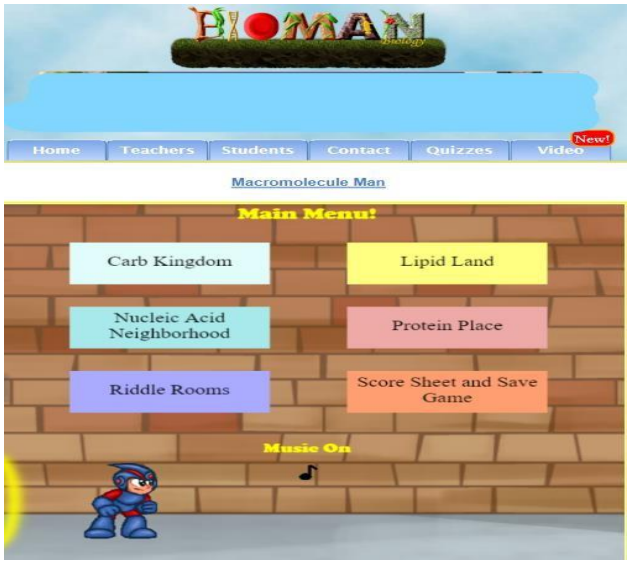


Figure 4. *BioMan Biology Virtual Simulation*

MONITORING METACOGNITIVE CHECKPOINT

DIRECTION: Answer and brainstorm the following questions with your groupmates.

- Do we understand what we are doing?
- Does the task make sense?
- Are we reaching our goals?
- Are the strategies working well to complete the task within the given time limit? If not, what should we do?
- Do we need to make improvements?



JUSTIFYING ARGUMENTS

DIRECTION: After exploring the simulation, finalize your work and data. Record your observations on the digital argumentation board using the template in Canva by clicking on this link: http://tinyurl.com/W3ADIBoard or using a QR scanner application, scan the QR code. To answer the guiding question, you need to develop a scientific argument consisting of the following:

SCAN ME

- **claim** (an explicit statement or conclusion which answers the guiding question),
- **evidence** (data observed and collected from the exploration that supports your claim), and
- **reasoning** (explains the importance of data presented to support your claim and connects related scientific principles to provide evidence to your claim).

To share your initial findings with others, we will employ a round-robin arrangement. This implies that two members from your group will remain stationary at your room table (RT) station to explain your group's work while the remaining members of your group will round to the remaining RT stations consecutively to attentively perceive and evaluate the initial findings formulated by your classmates.

Be ready to present your group's initial scientific argument. The aim of the argumentation session is not to convince others that your argument is the best, but the goal is to identify errors or faulty reasoning in the arguments and fix those mistakes. You will need to evaluate the content of the claim, the quality of the evidence, and the strength of the reasoning in each argument. To share your argument, you may raise your hand as your teacher facilitates the discussion.

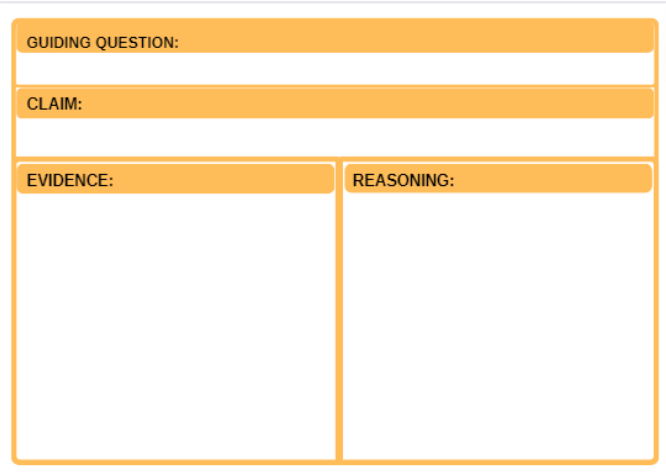


Figure 5. *Initial Argumentation Board*

EXPOUNDING ERA

DIRECTION: Using your web browser, visit the link: http://tinyurl.com/BiomoleculesDiscussion or using a QR scanner application, scan the QR code below for you to access BYJU'S website for further elaboration of the topic. Sample problem-solving items and real-life examples are also on the website.

SCAN ME

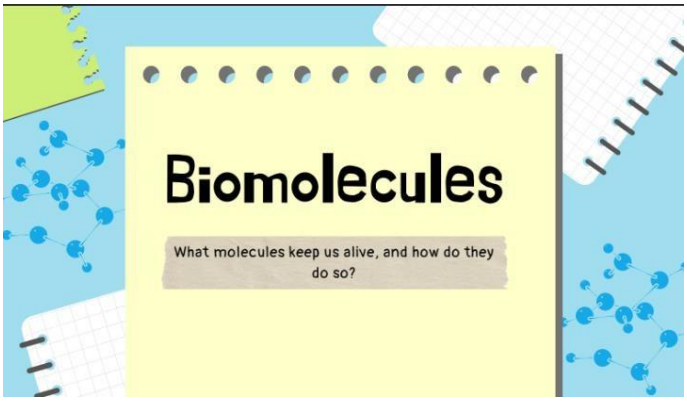


Figure 6. *Discussion of Biomolecules PDF*

EVALUATING METACOGNITIVE CHECKPOINT



DIRECTION: Answer and brainstorm the following questions with your groupmates.

- Have we reached our goal?
- What strategy worked?
- What strategy did not work?
- Would we do things differently next time?

CHECKING COMPOSITIONS

DIRECTION: Once you have completed your group investigation, participated in the argumentation session, and reflective post-discussion, you will need to finalize your individual written scientific argumentation report. Be reminded that your scientific argument must consist of a claim (a clear statement that answers the guiding question), **evidence** (data from the investigation that supports your claim), and **reasoning** (explains clearly why the data you presented supports your claim and consists of underlying scientific principles that link evidence to your claim).

GUIDING QUESTION:	
CLAIM:	
EVIDENCE:	REASONING:

REFLECTIVE LOG

DIRECTION: You may now answer the following reflective questions via Google Forms by clicking on this link: http://tinyurl.com/W3REFLECTIVELOG or scanning the QR code below.

SCAN ME

1. What were the key concepts covered in today's lesson?

_____

_____

_____

2. How can I apply the knowledge gained from this lesson in real-life situations?

_____

_____

_____

3. If there was feedback provided by the instructor or peers, what insights can I gain from it?

_____

_____

_____

4. Did I ask questions when I needed clarification?

_____

_____

_____

LESSON 4

LAW OF CONSERVATION OF MASS

LEARNING OBJECTIVES:

At the end of the lesson, the learners must be able to:

- Define the Law of Conservation of Mass;
- Investigate upon the Law of Conservation of Mass using a virtual simulation; and
- Write reflection about the importance of chemical reactions in practical scenarios such as
- combustion.

DURATION:

4 Hours

INTRODUCTION:

Ever misplace a screw? The following scenario occurs far too frequently. To clean up an apparatus, you have disassembled it. Upon reassembling the apparatus, it seems like you have one or more extra screws. Or you discover that one of the original equipment's screws is missing. Either way, you are aware that something is off. You anticipate finishing up with the same quantity of material—neither more nor less than what you began with.

Welcome to the fascinating world of the Law of Conservation of Mass! Imagine a cosmic dance where atoms gracefully swap partners, molecules perform intricate choreography, yet the grandeur of the total mass remains an unwavering constant.

CHEM-O-TRIVIA

Did you know? The Law of Conservation of Mass applies to all types of chemical reactions and is in accordance with Einstein's theory of relativity.

REFRESHING KNOWLEDGE

DIRECTION: Using your web browser, visit the link: http://tinyurl.com/W4RefreshingKnowledge to access the Word Wall website, or using a QR code scanner app, scan the QR code below. Click the start button and play the Anagram game. Then upload a screenshot or picture of your garnered score on the g-classroom "W4 Refreshing Knowledge" classwork.

SCAN ME



Figure 1. *Interface of Anagram in Word Wall*

GUIDING QUESTION

In this lesson, you will be formulating a scientific argument to answer the guiding question:

How is mass being conserved after a chemical reaction?

PLANNING METACOGNITIVE CHECKPOINT

DIRECTION: Answer and brainstorm the following questions with your groupmates.

- What is the description of the task?
- Why this task is important to us?
- What is our main goal?
- What are the resources that we need to successfully finish this task?
- How much time will we need to finish this task?
- How do we divide the task?



INVESTIGATING MOMENTS

DIRECTION: Using your web browser, visit the link: http://tinyurl.com/W4VirtualSimulation to access Olabs Virtual Simulation, or using a QR code scanner application, scan the QR code below. Working collaboratively with your groupmates, feel free to explore the simulation to record your observations and gather relevant data that will serve as evidence in your initial scientific argument.

SCAN ME

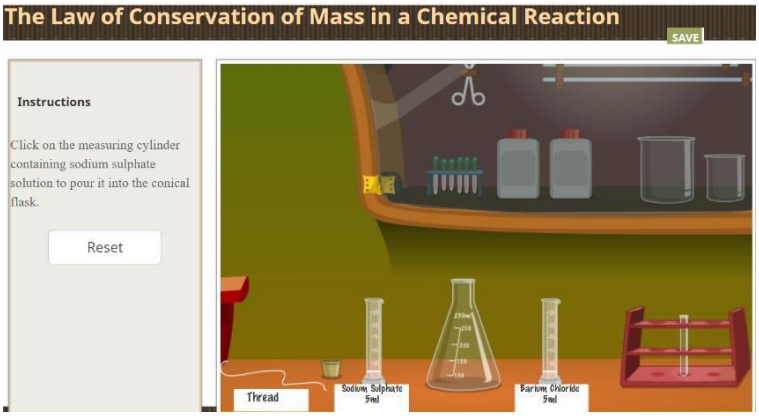


Figure 4. *Olabs Virtual Simulation*

MONITORING METACOGNITIVE CHECKPOINT

DIRECTION: Answer and brainstorm the following questions with your groupmates.

- Do we understand what we are doing?
- Does the task make sense?
- Are we reaching our goals?
- Are the strategies working well to complete the task within the given time limit? If not, what should we do?
- Do we need to make improvements?



JUSTIFYING ARGUMENTS

DIRECTION: After exploring the simulation, finalize your work and data. Record your observations on the digital argumentation board using the template in Canva by clicking on this link: http://tinyurl.com/W4ADIBoard or using a QR scanner application, scan the QR code. To answer the guiding question, you need to develop a scientific argument consisting of the following:

SCAN ME

- **claim** (an explicit statement or conclusion which answers the guiding question),
- **evidence** (data observed and collected from the exploration that supports your claim), and
- **reasoning** (explains the importance of data presented to support your claim and connects related scientific principles to provide evidence to your claim).

To share your initial findings with others, we will employ a round-robin arrangement. This implies that two members from your group will remain stationary at your room table (RT) station to explain your group's work while the remaining members of your group will round to the remaining RT stations consecutively to attentively perceive and evaluate the initial findings formulated by your classmates.

Be ready to present your group's initial scientific argument. The aim of the argumentation session is not to convince others that your argument is the best, but the goal is to identify errors or faulty reasoning in the arguments and fix those mistakes. You will need to evaluate the content of the claim, the quality of the evidence, and the strength of the reasoning in each argument. To share your argument, you may raise your hand as your teacher facilitates the discussion.

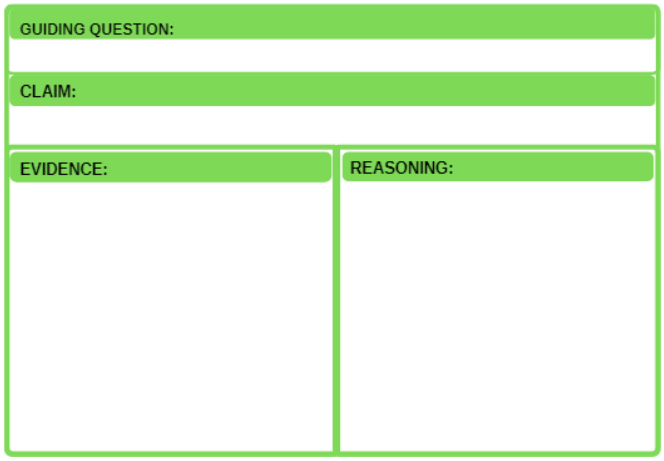


Figure 5. *Initial Argumentation Board*

EXPOUNDING ERA

DIRECTION: Using your web browser, visit the link: http://tinyurl.com/W4Filscihub or using a QR scanner application, scan the QR code below for you to access Filipino Science Hub website for further elaboration of the topic. Sample problem-solving items and real-life examples are also on the website.

SCAN ME

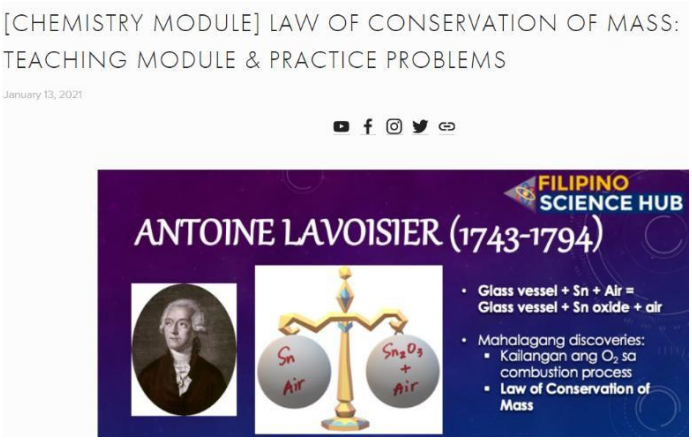


Figure 6. *Discussion about Law of Conservation of Mass*

EVALUATING METACOGNITIVE CHECKPOINT



DIRECTION: Answer and brainstorm the following questions with your groupmates.

- Have we reached our goal?
- What strategy worked?
- What strategy did not work?
- Would we do things differently next time?

CHECKING COMPOSITIONS

DIRECTION: Once you have completed your group investigation, participated in the argumentation session, and reflective post-discussion, you will need to finalize your individual written scientific argumentation report. Be reminded that your scientific argument must consist of a claim (a clear statement that answers the guiding question), **evidence** (data from the investigation that supports your claim), and **reasoning** (explains clearly why the data you presented supports your claim and consists of underlying scientific principles that link evidence to your claim).

GUIDING QUESTION:	
CLAIM:	
EVIDENCE:	REASONING:

REFLECTIVE LOG

DIRECTION: You may now answer the following reflective questions via Google Forms by clicking on this link: http://tinyurl.com/W4REFLECTIVELOG or scanning the QR code below.

SCAN ME

1. What were the key concepts covered in today's lesson?

   _____

   _____

   _____

2. What aspects of the lesson were challenging for me?

   _____

   _____

   _____

3. How will you apply your learnings in real-life situations?

   _____

   _____

   _____

4. Based on today's lesson, what specific goals do I have for improvement?

   _____

   _____

   _____

LESSON 5
BALANCING CHEMICAL EQUATIONS

LEARNING OBJECTIVES:

At the end of the lesson, the learners must be able to:

- Explain the factors required for a chemical equation to be balanced;
- Perform an investigation about chemical reactions using a virtual simulation; and
- Summarize the importance of balancing chemical equations.

DURATION:

4 Hours

INTRODUCTION:

Balancing chemical equations is not just a scientific act; it's a poetic dance of numbers, a choreography of atoms that ensures the cosmic ballet maintains its graceful equilibrium. As you embark on this journey, you become the alchemist, transforming chaos into order, and unlocking the secrets of matter's elegant choreography.

Atoms in the reactants do not vanish, and new atoms do not appear to form the products, even though chemical compounds are broken up and new compounds are generated during a chemical reaction. Atoms do not make or destroy themselves during chemical processes. The products include the same atoms as the reactants; they have just been rearranged into various configurations. The reactant and product sides of a chemical equation need to include the same number of atoms for the equation to be complete.

CHEM-O-TRIVIA Did you know? Life itself is a grand balancing act of chemical equations – from the metabolic reactions within our cells to the delicate equilibrium of ecosystems.

REFRESHING KNOWLEDGE

DIRECTION: Using your web browser, visit the link: http://tinyurl.com/W5RefreshingKnowledge to access the Word Wall website, or using a QR code scanner app, scan the QR code below. Click the start button and play the Balancing Chemical Equation Quiz game. Then upload a screenshot or picture of your garnered score on the g-classroom "W5 Refreshing Knowledge" classwork.

SCAN ME

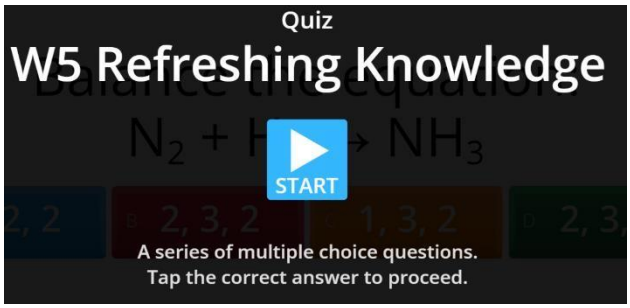


Figure 1. *Interface of Balancing Chemical Equation Quiz in Word Wall*

GUIDING QUESTION

In this lesson, you will be formulating a scientific argument to answer the guiding question:

What is the balanced chemical equation for the combustion of methane? How might the methane's chemical equation provide insight on its effects on the environment?

PLANNING METACOGNITIVE CHECKPOINT

DIRECTION: Answer and brainstorm the following questions with your groupmates.

- What is the description of the task?
- Why this task is important to us?
- What is our main goal?
- What are the resources that we need to successfully finish this task?
- How much time will we need to finish this task?
- How do we divide the task?



INVESTIGATING MOMENTS

DIRECTION: Using your web browser, visit the link: http://tinyurl.com/W5VirtualSimulation to access PhET Virtual Simulation, or using a QR code scanner application, scan the QR code below. To start the simulation, click on the "Introduction" button. Working collaboratively with your groupmates, feel free to explore the simulation to record your observations and gather relevant data that will serve as evidence in your initial scientific argument.

SCAN ME

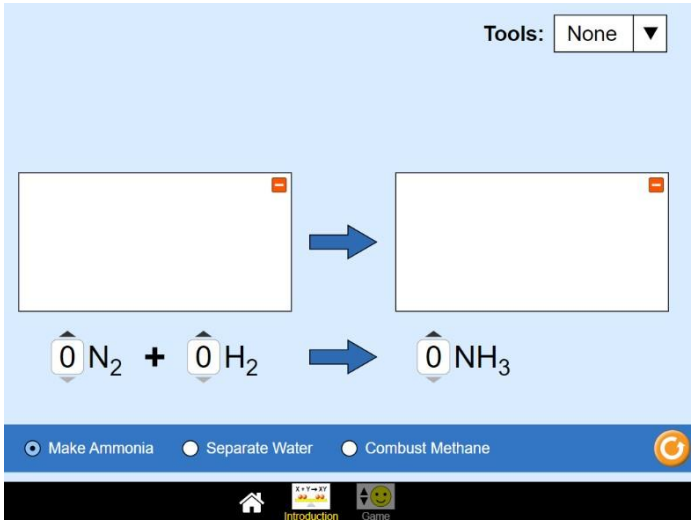


Figure 4. *PhET Virtual Simulation*

MONITORING METACOGNITIVE CHECKPOINT

DIRECTION: Answer and brainstorm the following questions with your groupmates.

- Do we understand what we are doing?
- Does the task make sense?
- Are we reaching our goals?
- Are the strategies working well to complete the task within the given time limit? If not, what should we do?
- Do we need to make improvements?



JUSTIFYING ARGUMENTS

DIRECTION: After exploring the simulation, finalize your work and data. Record your observations on the digital argumentation board using the template in Canva by clicking on this link: http://tinyurl.com/W5ADIBoard or using a QR scanner application, scan the QR code. To answer the guiding question, you need to develop a scientific argument consisting of the following:

SCAN ME

- **claim** (an explicit statement or conclusion which answers the guiding question),
- **evidence** (data observed and collected from the exploration that supports your claim), and
- **reasoning** (explains the importance of data presented to support your claim and connects related scientific principles to provide evidence to your claim).

To share your initial findings with others, we will employ a round-robin arrangement. This implies that two members from your group will remain stationary at your room table (RT) station to explain your group's work while the remaining members of your group will round to the remaining RT stations consecutively to attentively perceive and evaluate the initial findings formulated by your classmates.

Be ready to present your group's initial scientific argument. The aim of the argumentation session is not to convince others that your argument is the best, but the goal is to identify errors or faulty reasoning in the arguments and fix those mistakes. You will need to evaluate the content of the claim, the quality of the evidence, and the strength of the reasoning in each argument. To share your argument, you may raise your hand as your teacher facilitates the discussion.

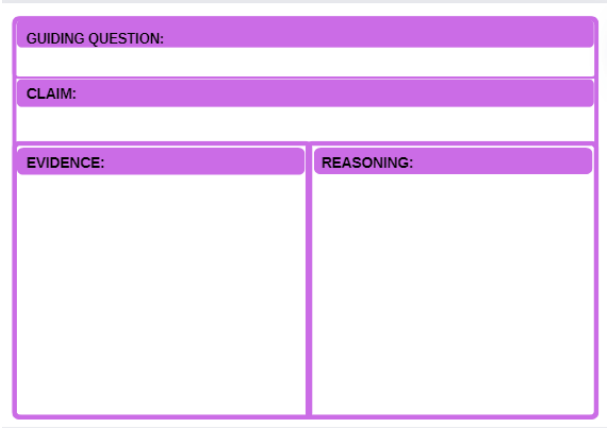


Figure 5. *Initial Argumentation Board*

EXPOUNDING ERA

DIRECTION: Using your web browser, visit the link: http://tinyurl.com/W5Filscihub or using a QR scanner application, scan the QR code below for you to access Filipino Science Hub website for further elaboration of the topic. Sample problem-solving items and real-life examples are also on the website.

SCAN ME

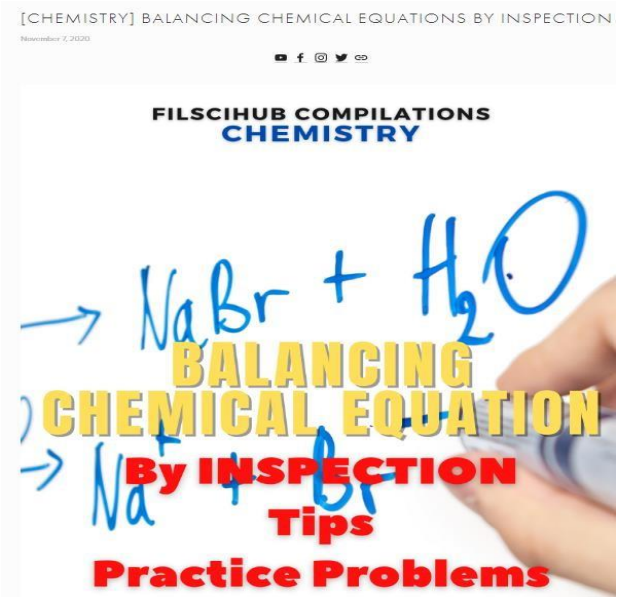


Figure 6. *Discussion about Law of Conservation of Mass*

EVALUATING METACOGNITIVE CHECKPOINT



DIRECTION: Answer and brainstorm the following questions with your groupmates.

- Have we reached our goal?
- What strategy worked?
- What strategy did not work?
- Would we do things differently next time?

CHECKING COMPOSITIONS

DIRECTION: Once you have completed your group investigation, participated in the argumentation session, and reflective post-discussion, you will need to finalize your individual written scientific argumentation report. Be reminded that your scientific argument must consist of a claim (a clear statement that answers the guiding question), **evidence** (data from the investigation that supports your claim), and **reasoning** (explains clearly why the data you presented supports your claim and consists of underlying scientific principles that link evidence to your claim).

GUIDING QUESTION:	
CLAIM:	
EVIDENCE:	REASONING:

REFLECTIVE LOG

DIRECTION: You may now answer the following reflective questions via Google Forms by clicking on this link: http://tinyurl.com/W5REFLECTIVELOG or scanning the QR code below.

SCAN ME

1. Which helps you to learn more in this session?

   _____

   _____

   _____

2. How does this lesson connect with previous knowledge or topics?

   _____

   _____

   _____

3. What are the main takeaways or key points I want to remember from this lesson?

   _____

   _____

   _____

4. Do I need to adjust my study strategies based on today's experience?

   _____

   _____

   _____

LESSON 6
FACTORS AFFECTING REACTION RATES

LEARNING OBJECTIVES:

At the end of the lesson, the learners must be able to:

- Describe each factors affecting chemical reactions
- Examine the concepts of collision theory using a virtual simulation;
- Reflect how the factors affecting rates of chemical reactions are utilized in food industry and materials production, control of fire, pollution, and corrosion;

DURATION:

4 Hours

INTRODUCTION:

Welcome to the amazing realm of chemical reactions! In the field of chemistry, a variety of factors influence how compounds change. Deciphering these elements is essential to solving the puzzles of how and why responses take place. As we delve into the complexities of chemical reactions and reveal the mysteries that underpin the fascinating field of chemistry, join us on a trip to examine the dynamic interplay of these components. Chemical reactions can occur at a wide range of rates for the consumption of reactants and the formation of products. The chemical makeup of the reacting substances, the reactants' particle size, concentration, temperature, and the presence of a catalyst are the five variables that can be found to influence the rates of chemical reactions.

CHEM-O-TRIVIA Did you know? The area of chemistry that examines reaction rates is called chemical kinetics. It include calculating rate laws, measuring and analyzing reaction rates, and looking into the variables that influence reaction rates.

REFRESHING KNOWLEDGE

DIRECTION: Using your web browser, visit the link: http://tinyurl.com/W6RefreshingKnowledge to access the Word Wall website, or using a QR code scanner app, scan the QR code below. Click the start button and play the Crossword game. Then upload a screenshot or picture of your garnered score on the g-classroom "W6 Refreshing Knowledge" classwork.

SCAN ME



Figure 1. *Interface of Crossword in Word Wall*

GUIDING QUESTION

In this lesson, you will be formulating a scientific argument to answer the guiding question:

How does change in temperature affect the rate of collision of molecules?

PLANNING METACOGNITIVE CHECKPOINT

DIRECTION: Answer and brainstorm the following questions with your groupmates.

- What is the description of the task?
- Why this task is important to us?
- What is our main goal?
- What are the resources that we need to successfully finish this task?
- How much time will we need to finish this task?
- How do we divide the task?



INVESTIGATING MOMENTS

DIRECTION: Using your web browser, visit the link: http://tinyurl.com/W6VirtualSimulation to access PhET Virtual Simulation, or using a QR code scanner application, scan the QR code below. To start the simulation, click on the "Many Collisions" tab. Working collaboratively with your groupmates, feel free to explore the simulation to record your observations and gather relevant data that will serve as evidence in your initial scientific argument.

SCAN ME

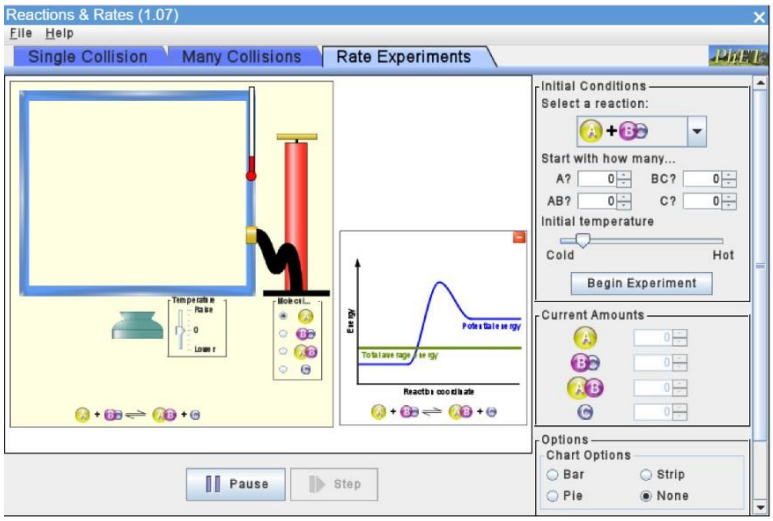


Figure 4. *PhET Virtual Simulation*

MONITORING METACOGNITIVE CHECKPOINT

DIRECTION: Answer and brainstorm the following questions with your groupmates.

- Do we understand what we are doing?
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- Are the strategies working well to complete the task within the given time limit? If not, what should we do?
- Do we need to make improvements?



JUSTIFYING ARGUMENTS

DIRECTION: After exploring the simulation, finalize your work and data. Record your observations on the digital argumentation board using the template in Canva by clicking on this link: http://tinyurl.com/W6ADIBoard or using a QR scanner application, scan the QR code. To answer the guiding question, you need to develop a scientific argument consisting of the following:

SCAN ME

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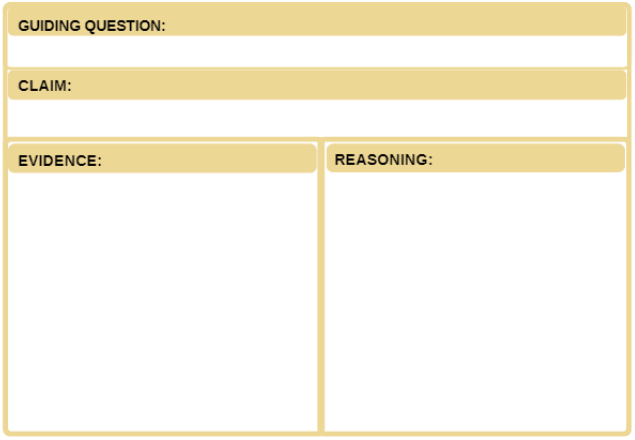


Figure 5. *Initial Argumentation Board*

EXPOUNDING ERA

DIRECTION: Using your web browser, visit the link: http://tinyurl.com/W6LumenLearning or using a QR scanner application, scan the QR code below for you to access Lumen Learning website for further elaboration of the topic. Sample problem-solving items and real-life examples are also on the website.

SCAN ME

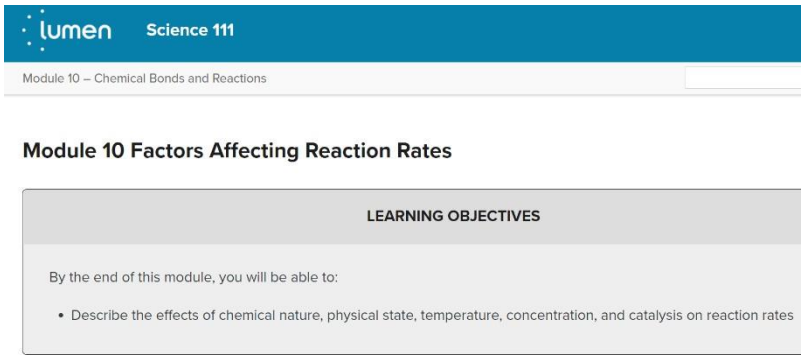


Figure 6. *Discussion about Factors affecting Reaction Rates*

EVALUATING METACOGNITIVE CHECKPOINT



DIRECTION: Answer and brainstorm the following questions with your groupmates.

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- What strategy worked?
- What strategy did not work?
- Would we do things differently next time?

CHECKING COMPOSITIONS

DIRECTION: Once you have completed your group investigation, participated in the argumentation session, and reflective post-discussion, you will need to finalize your individual written scientific argumentation report. Be reminded that your scientific argument must consist of a claim (a clear statement that answers the guiding question), **evidence** (data from the investigation that supports your claim), and **reasoning** (explains clearly why the data you presented supports your claim and consists of underlying scientific principles that link evidence to your claim).

GUIDING QUESTION:	
CLAIM:	
EVIDENCE:	REASONING:

REFLECTIVE LOG

DIRECTION: You may now answer the following reflective questions via Google Forms by clicking on this link: http://tinyurl.com/W6REFLECTIVELOG or scanning the QR code below.

SCAN ME

1. What are the three important things have you learned in the session?

   _____

   _____

   _____

2. What is one thing that you want to share with others?

   _____

   _____

   _____

3. How will you apply your learnings in real-life situations?

   _____

   _____

   _____

4. Were the explanations and examples provided by the instructor clear and easy to follow?

   _____

   _____

   _____

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